Bidisha Das

Senior Software Engineer · Backend & Platform Engineering · AI Infrastructure · Enterprise Authorization

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SUMMARY

Backend and platform engineer with 6 years of production experience, currently a Senior Member of Technical Staff at Salesforce. I build enterprise authorization and AI-agent infrastructure: Java and Python services, metadata-grounded agent workflows, MCP integrations, and permission-aware tool execution behind human approval, with authorization decisions kept in deterministic code. Earlier: playback reliability at Disney+ Hotstar, fintech journeys at MakeMyTrip, and platform tooling at AutoRABIT.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript/JavaScript, SQL

Backend & APIs: Spring Boot, FastAPI, REST, Salesforce Metadata and Tooling APIs, GitHub REST API, OData v4

Authorization & safety: RBAC and enterprise access management, permission-aware tool execution, approval-gated writes, OAuth, fail-closed checks

AI infrastructure: MCP servers, LangGraph (interrupt/resume, Postgres checkpointing), RAG with Pinecone, agent evaluation (rubric-based LLM judge, must-pass safety cases)

Data & infrastructure: PostgreSQL, Docker, Kubernetes, AWS, GitHub Actions CI, Splunk

Frontend (supporting): React, Next.js, Angular, Vue, Shaka Player (DASH/HLS)

EXPERIENCE

Salesforce

Dec 2024 – Present

Senior Member of Technical Staff · Bengaluru, India

- Designed, built and deployed **five production AI agent workflows for enterprise access management**, cutting a ~60-day manual process to ~5 days of self-serve execution (~92% faster).
- Extended the in-house Agentforce platform with a **retrieval-to-action layer**: natural-language requests retrieve versioned Skills, enterprise metadata grounds the plan, and permission-aware tools execute structured operations; 10+ teams self-serve on it.
- **Authored the MCP server** that exposes eight enterprise capabilities to agents, with structured tool contracts, explicit failure semantics and permission-aware execution.
- Shipped a **Slack agent adopted by 150+ people across 8 teams**, mostly non-engineers; privileged operations run only under the requester's own permissions and behind human approval.
- Built governed **Metadata API and Tooling API integrations** across roles, groups, queues, sharing rules and access policies, with schema validation and authorization decisions in deterministic code, reducing misconfigured access requests by ~35%.
- Built **Java services on PostgreSQL** for role and queue access summaries, restriction rules and public-group membership queries, plus field-level security checks.
- Built the **agent evaluation and regression framework**: behavioral checks on retrieval, tool selection and permissions; an LLM judge with a written rubric and pass thresholds; permission safety cases that must pass on every change.
- Migrated five workflows from a legacy V1 retrieval-driven architecture to the redesigned V2 platform, consolidating 20+ prompts, permissions, tool definitions and structured outputs.
- Deployed and operated agent services on AWS with Docker and Kubernetes; event-driven integrations decoupled workflows from downstream systems, improving throughput by ~40%. Mentor 5+ engineers through code and design reviews.

Disney+ Hotstar

Jan 2023 – Dec 2024

Software Development Engineer II · Playback (web and TV) · Remote

- Reduced player failure rate by **~70%** by matching recovery to each error class: bounded auto-retry, failover to a secondary source and variant blacklisting, without restarting playback.
- Fixed session analytics that counted retries as new sessions by redesigning playback-session identity; defined startup, rebuffer and failure metrics with Product and Data Science.
- Cut load time by **~10% for 50M+ users** with segment prefetching and a concurrency-capped preload manager that parses the manifest while DRM initializes in parallel.

- Built a web-worker-based playback reporting API for real-time error tracking, and caption sync for stitched ads on Apple HLS across 50+ countries.

MakeMyTrip Jun 2022 – Jan 2023

Senior Software Engineer · TripMoney (fintech) · Bengaluru, India

- Built lending, credit-card and forex journeys and integrated partner APIs with retry and validation, raising transaction success by 18%, cutting API latency by ~30% and lowering application drop-off by ~25%.
- Led web performance work (LCP, FCP, TTFB) through code-splitting, image compression and bundle-size reduction.

AutoRABIT Sep 2020 – Jun 2022

Software Developer · Hyderabad, India

- Built a connector that loads external CSV data into Salesforce using Java, Spring Boot and Apache Olingo (OData v4), hosted on AWS EC2 and S3.
- Built the organisation-wide design system (Angular, PrimeNG, NgRx, Web Components) in a micro-frontend setup, used as the single source of truth for UI.
- Delivered a Configure-Price-Quote (CPQ) application that contributed to ~$2M in revenue.

PERSONAL PROJECTS

ResolveFlow · GitHub issue diagnosis with approval-gated execution github.com/officialbidisha/ResolveFlow

- LangGraph agent that diagnoses GitHub issues with RAG citations over ~2,750 closed-issue chunks; an independent LLM review whose approve/escalate decision is computed in code. FastAPI + React, deployed on Render and Vercel.
- Human approval is a LangGraph `interrupt()` checkpointed in PostgreSQL and resumable from a later request. Execution refuses without recorded approval, the only write tools are comment and label, and writes run under the requester's GitHub OAuth token with a Postgres write ledger for retry-safe posting.
- GitHub Actions runs 13 code-graded safety-gate checks (100% pass bar) and the approval-path and concurrency test suites on every pull request; threshold-gated capability evals on real issues run on main.

BhashaRoute · Multilingual intent router with a safety gate github.com/officialbidisha/bhasharoute

- English/Hindi/Bengali intent routing with calibrated abstention; 87.8% macro-F1 vs 62.0% for keyword rules on a human-translated test set of 8,157 requests, and a fail-closed gate with 99.5% recall on state-changing requests (Python, scikit-learn, FastAPI).

OPEN SOURCE

Shaka Player (Google's DASH/HLS player): merged subtitle-role fixes for HLS (#5357) and DASH role/kind="subtitle" with unit tests (#10441).

EDUCATION

St. Thomas' College of Engineering & Technology (MAKAUT) 2016 – 2020

B.Tech, Computer Science & Engineering · CGPA 9.29/10